

ELECTRIC VEHICLES, BATTERIES AND ALTERNATIVE FUELS

POWERTRAIN SOURCE-OF-ELECTRICITY MAP → CELL TO PACK SYSTEMS → LITHIUM-ION DISCHARGE RAIL → BATTERY CHEMISTRY DECISION MATRIX → ENERGY AND POWER AXES → AC AND DC CHARGING

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 POWERTRAIN SOURCE-OF-ELECTRICITY MAP

POWERTRAIN SOURCE-OF-ELECTRICITY MAP BATTERY STORES ALL TRACTION ELECTRICITY ENGINE + REGENERATIVE BRAKING CHARGE NON-PLUG-IN BATTERY EXTERNAL PLUG + ENGINE BACKUP HYDROGEN FUEL CELL GENERATES ONBOARD ELECTRICITY WHERE DOES TRACTION ELECTRICITY ORIGINATE?

BEV → battery stores all traction electricity

HEV → engine + regenerative braking charge non-plug-in battery

PHEV → external plug + engine backup

FCEV → hydrogen fuel cell generates onboard electricity

CONCEPT / ARCHITECTURE

- BEV → battery stores all traction electricity
- HEV → engine + regenerative braking charge non-plug-in battery

MECHANISM / APPLICATION

- PHEV → external plug + engine backup
- FCEV → hydrogen fuel cell generates onboard electricity

READINESS / STATUS LIMIT

- TEST → where does traction electricity originate?
- BEV → battery stores all traction electricity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FCEV → hydrogen fuel cell generates onboard electricity.

01

STAGE 01 CELL TO PACK SYSTEMS HIERARCHY

CELL TO PACK SYSTEMS HIERARCHY ELECTROCHEMICAL UNIT GROUPED CELLS + MECHANICAL INTEGRATION MODULES/CELLS + ENCLOSURE + BUSBARS + SENSING BMS + CONTACTORS + PROTECTION COOLING/HEATING + PROPAGATION BARRIERS

CELL → electrochemical unit

MODULE → grouped cells + mechanical integration

PACK → modules/cells + enclosure + busbars + sensing

CONTROL → BMS + contactors + protection

CONCEPT / ARCHITECTURE

- CELL → electrochemical unit
- MODULE → grouped cells + mechanical integration

MECHANISM / APPLICATION

- PACK → modules/cells + enclosure + busbars + sensing
- CONTROL → BMS + contactors + protection

READINESS / STATUS LIMIT

- THERMAL → cooling/heating + propagation barriers
- CELL → electrochemical unit

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PACK → modules/cells + enclosure + busbars + sensing.

02

STAGE 02 LITHIUM-ION DISCHARGE RAIL

LITHIUM-ION DISCHARGE RAIL RELEASES LI+ DURING DISCHARGE CARRIES IONS BLOCKS DIRECT ELECTRONIC CONTACT RECEIVES LI+ DURING DISCHARGE EXTERNAL CIRCUIT ELECTRONS POWER MOTOR

ANODE → releases Li+ during discharge

ELECTROLYTE → carries ions

SEPARATOR → blocks direct electronic contact

CATHODE → receives Li+ during discharge

EXTERNAL CIRCUIT → electrons power motor

STARTING CONDITION / INPUT

- ANODE → releases Li+ during discharge
- ELECTROLYTE → carries ions

SCIENTIFIC OR ENGINEERING MECHANISM

- SEPARATOR → blocks direct electronic contact
- CATHODE → receives Li+ during discharge

OUTPUT / FAILURE BOUNDARY

- EXTERNAL CIRCUIT → electrons power motor
- ANODE → releases Li+ during discharge

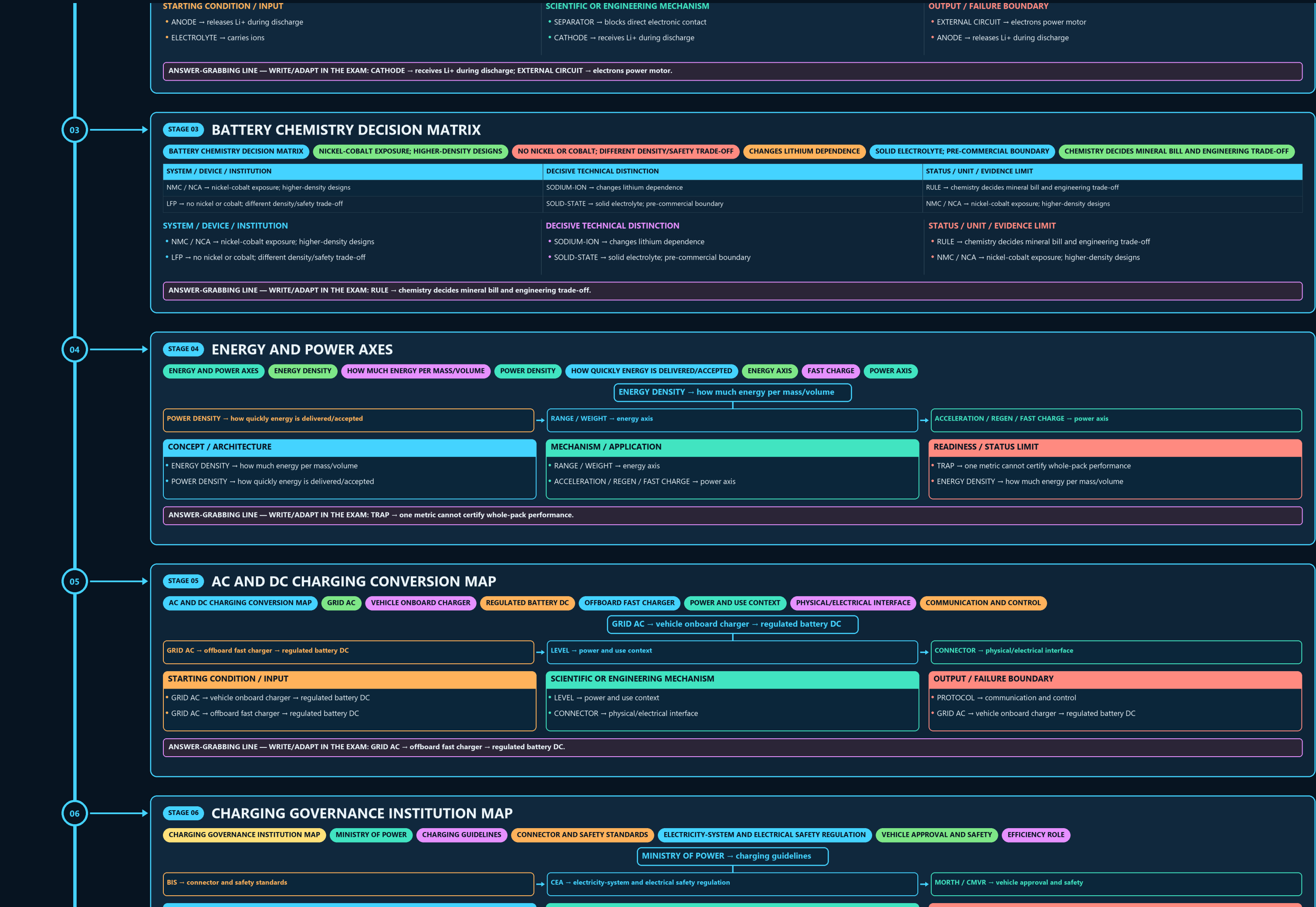
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CATHODE → receives Li+ during discharge; EXTERNAL CIRCUIT → electrons power motor.

03

STAGE 03 BATTERY CHEMISTRY DECISION MATRIX

BATTERY CHEMISTRY DECISION MATRIX NICKEL-COBALT EXPOSURE; HIGHER-DENSITY DESIGNS NO NICKEL OR COBALT; DIFFERENT DENSITY/SAFETY TRADE-OFF CHANGES LITHIUM DEPENDENCE SOLID ELECTROLYTE; PRE-COMMERCIAL BOUNDARY CHEMISTRY DECIDES MINERAL BILL AND ENGINEERING TRADE-OFF

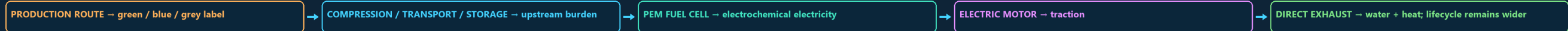
SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
NMC / NCA → nickel-cobalt exposure; higher-density designs	SODIUM-ION → changes lithium dependence	RULE → chemistry decides mineral bill and engineering trade-off
LFP → no nickel or cobalt; different density/safety trade-off	SOLID-STATE → solid electrolyte; pre-commercial boundary	NMC / NCA → nickel-cobalt exposure; higher-density designs



09

STAGE 09 HYDROGEN MOBILITY CHAIN

HYDROGEN MOBILITY CHAIN PRODUCTION ROUTE GREY LABEL UPSTREAM BURDEN PEM FUEL CELL ELECTROCHEMICAL ELECTRICITY ELECTRIC MOTOR DIRECT EXHAUST



STARTING CONDITION / INPUT

- PRODUCTION ROUTE → green / blue / grey label
- COMPRESSION / TRANSPORT / STORAGE → upstream burden

SCIENTIFIC OR ENGINEERING MECHANISM

- PEM FUEL CELL → electrochemical electricity
- ELECTRIC MOTOR → traction

OUTPUT / FAILURE BOUNDARY

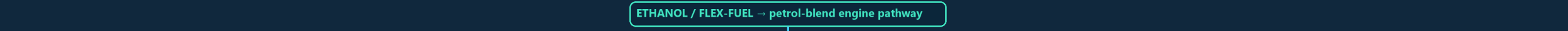
- DIRECT EXHAUST → water + heat; lifecycle remains wider
- PRODUCTION ROUTE → green / blue / grey label

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DIRECT EXHAUST → water + heat; lifecycle remains wider.

10

STAGE 10 ALTERNATIVE-FUEL PORTFOLIO

ALTERNATIVE-FUEL PORTFOLIO PETROL-BLEND ENGINE PATHWAY DIESEL-BLEND PATHWAY PURIFIED BIOGAS PATHWAY GASEOUS COMBUSTION FUELS HYDROGEN FUEL CELL ELECTROCHEMICAL ELECTRIC-DRIVE PATHWAY



CONCEPT / ARCHITECTURE

- ETHANOL / FLEX-FUEL → petrol-blend engine pathway
- BIODIESEL → diesel-blend pathway

MECHANISM / APPLICATION

- CBG / SATAT → purified biogas pathway
- CNG / LNG / HCNG → gaseous combustion fuels

READINESS / STATUS LIMIT

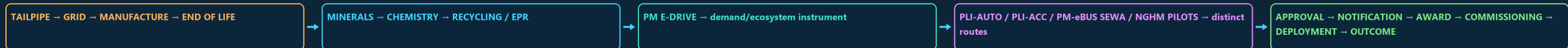
- HYDROGEN FUEL CELL → electrochemical electric-drive pathway
- ETHANOL / FLEX-FUEL → petrol-blend engine pathway

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CNG / LNG / HCNG → gaseous combustion fuels.

11

STAGE 11 INDIA TRANSITION AND EVIDENCE LADDER

INDIA TRANSITION AND EVIDENCE LADDER END OF LIFE PM E-DRIVE DEMAND/ECOSYSTEM INSTRUMENT PM-EBUS SEWA NGHM PILOTS DISTINCT ROUTES



STARTING CONDITION / INPUT

- TAILPIPE → GRID → MANUFACTURE → END OF LIFE
- MINERALS → CHEMISTRY → RECYCLING / EPR

SCIENTIFIC OR ENGINEERING MECHANISM

- PM E-DRIVE → demand/ecosystem instrument
- PLI-AUTO / PLI-ACC / PM-eBUS SEWA / NGHM PILOTS → distinct routes

OUTPUT / FAILURE BOUNDARY

- APPROVAL → NOTIFICATION → AWARD → COMMISSIONING → DEPLOYMENT → OUTCOME
- TAILPIPE → GRID → MANUFACTURE → END OF LIFE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: APPROVAL → NOTIFICATION → AWARD → COMMISSIONING → DEPLOYMENT → OUTCOME.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT THE QUESTION IS NOT “EV VERSUS NO EV”; IT BEVS USUALLY MAKE STRONGEST SENSE WHERE DAILY DRIVING RANGE HYDROGEN MOBILITY IS ANALYTICALLY STRONGER FOR HARD-TO-ELECTRIFY SEGMENTS THAN ROUTINE SMALL PERSONAL TRANSPORT, THOUGH COSTS AND INFRASTRUCTURE REMAIN ETHANOL/CNG/LNG MATTER BECAUSE THE MOBILITY TRANSITION IS BROADER THAN MINISTRY OF HEAVY INDUSTRIES PM E-DRIVE AND ACC BATTERY INDUSTRIAL-POLICY ANCHOR.

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- CAUTION: The question is not “EV versus no EV”; it is which powertrain fits which transport segment .
- CAUTION: BEVs usually make strongest sense where daily driving range is moderate and charging access can be organized.
- CAUTION: Hydrogen mobility is analytically stronger for hard-to-electrify segments than for
- routine small personal transport, though costs and infrastructure remain difficult.

READINESS, UNIT OR STATUS LIMIT

- CAUTION: Ethanol/CNG/LNG matter because the mobility transition is broader than battery electrification alone.
- Ministry of Heavy Industries: PM E-DRIVE and ACC battery industrial-policy anchor.
- PM E-DRIVE portal: official implementation interface and public scheme reference.
- ACC battery PLI framework: manufacturing-capacity building, not merely demand subsidy.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- Ministry of Power / BEE / BIS / CEA: charging guidelines, standards and power-system interface.
- MNRE: green-hydrogen mission and transport pilot guidelines.
- MoPNG: ethanol-blending and alternative-fuel ecosystem.
- EVs can reduce urban tailpipe emissions and fuel-import vulnerability in selected segments.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.